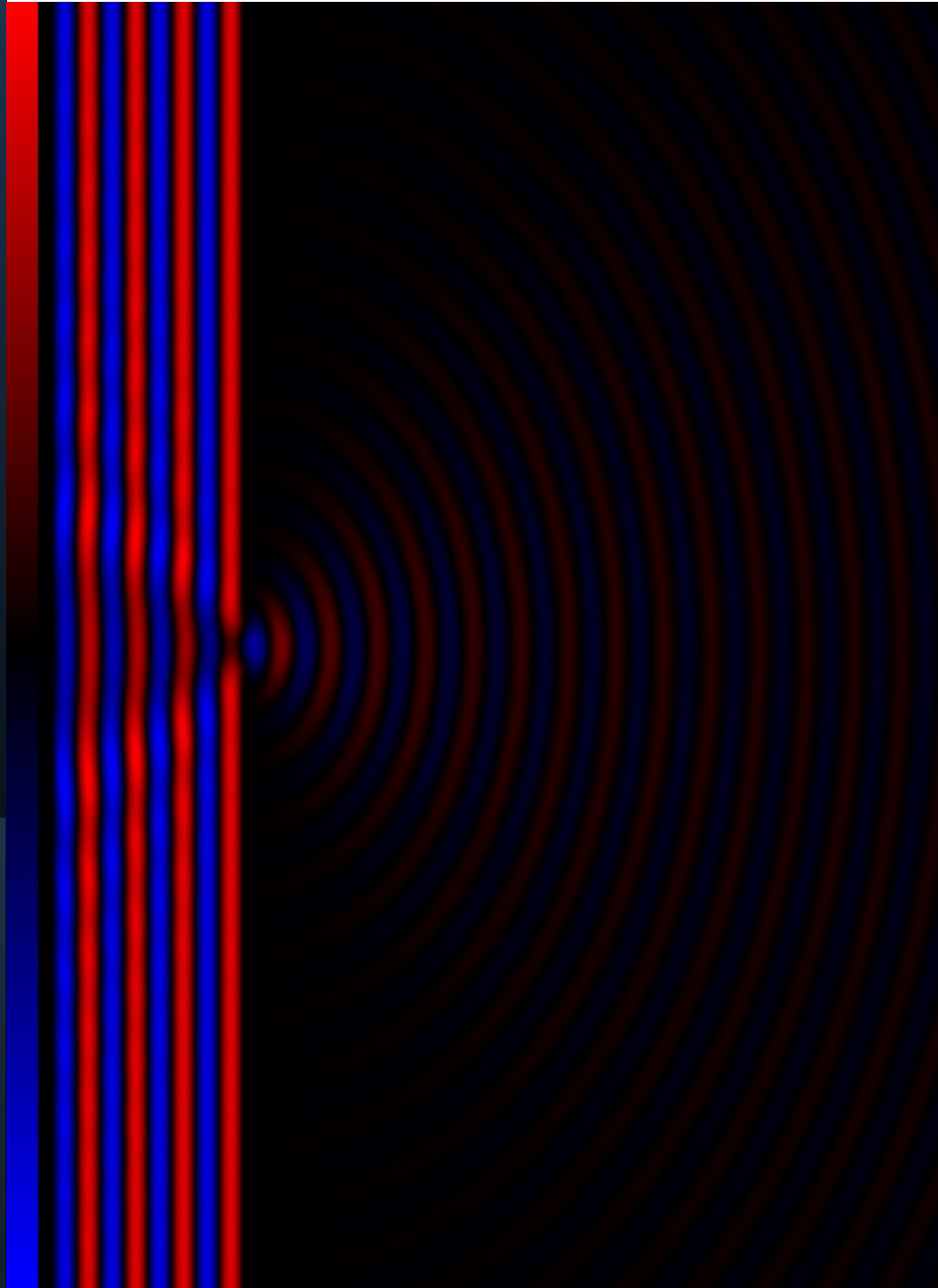
Diffraction

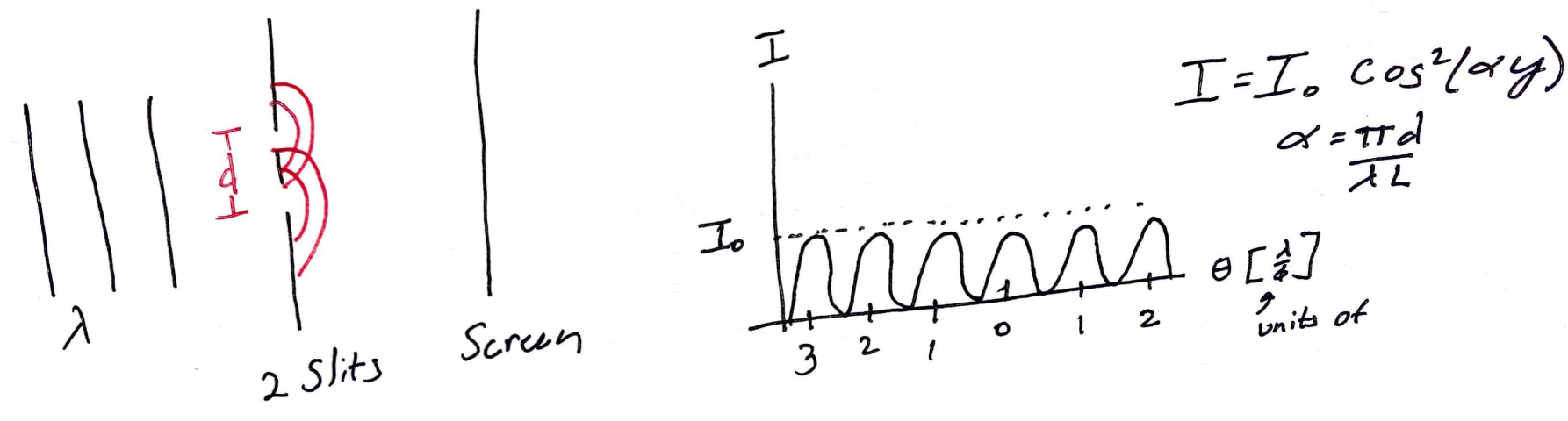


Review: Two Slit Interference Produces “Fringes”

1. In one dimension, assuming the to sources are coherent

$$d \sin\theta = m\lambda \quad \textbf{Constructive Int.}$$

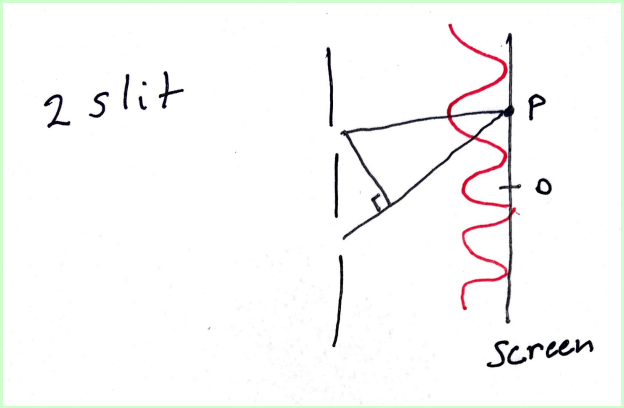
$$d \sin\theta x = \left(m + \frac{1}{2}\right)\lambda \quad \textbf{Destructive Int.}$$

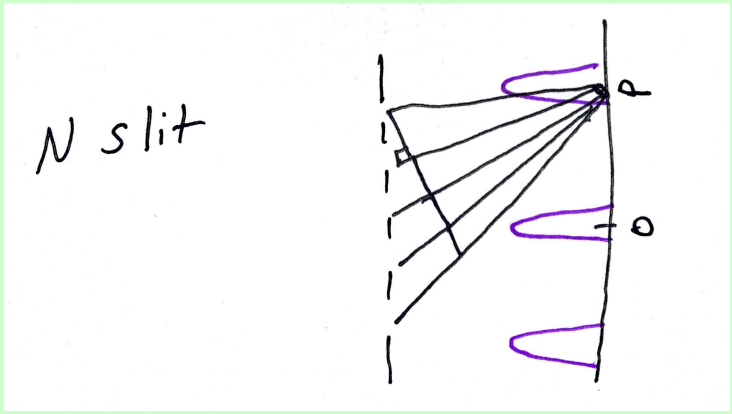


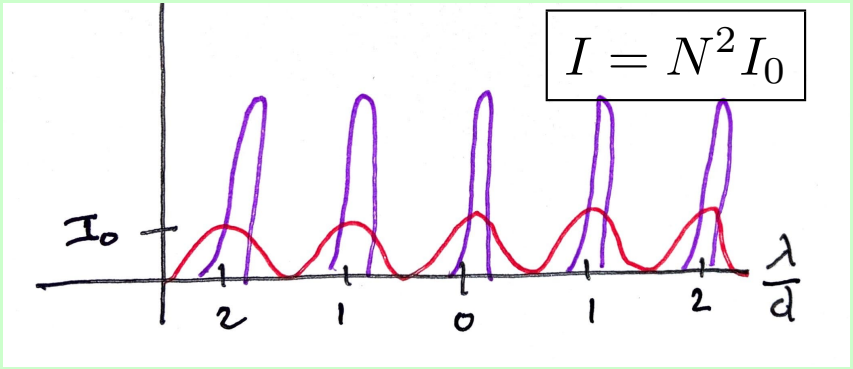
What happens when there are more than 2-slits? **Diffraction Gratings**

When there are more than two slits:

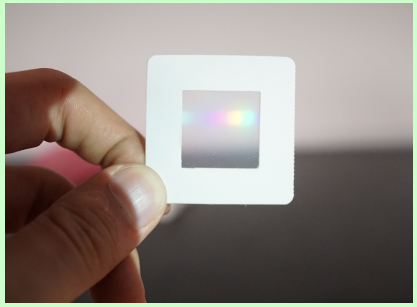
1. More sources contributing mean more intensity in peaks
2. Distant sources quickly get out of phase, leading to narrow peaks







Maxima still at $\sin\theta = m\frac{\lambda}{d}$



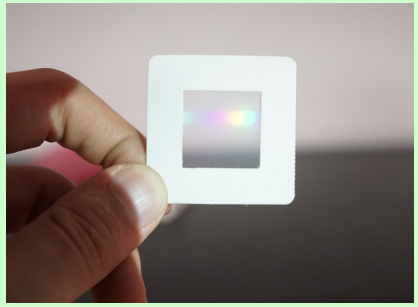
A diffraction grating with 500 lines/cm

Ex. A diffraction grating has 500 lines/ cm and is 3 cm long.

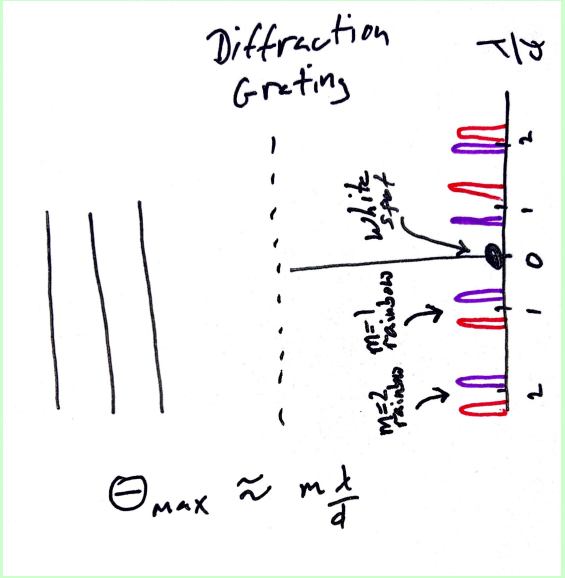
A. Find the distance between slits

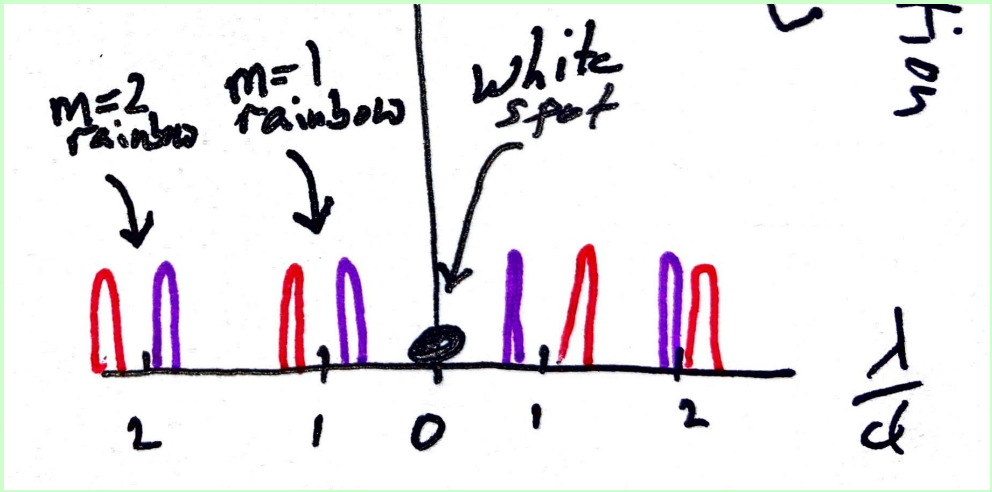
B. What are the total number of slits?

C. How much brighter is the central slits compared to I_0



Like prisms, diffraction gratings cause dispersion $\sin\theta = m\frac{\lambda}{d}$ or $\boxed{\theta \propto \lambda}$





Ex. A diffraction grating has 500 lines/ cm and is 1 cm long. White light is incident on the grating with wavelengths between 400 nm (violet) and 800 nm (red). The screen is L = 1.0 m away.

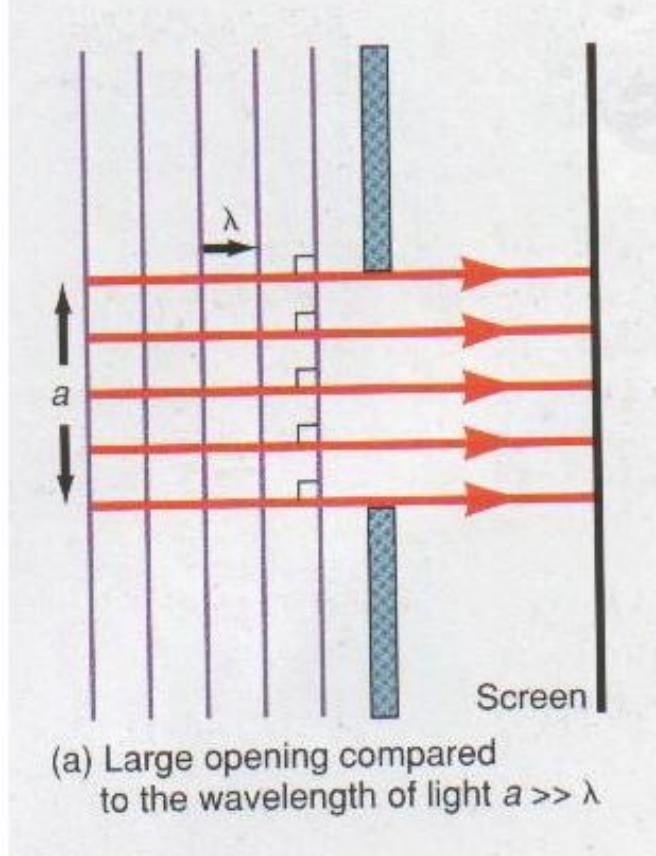
A. How long in cm is the spread the m = 1 rainbow?

B. How long in cm is the spread the m = 2 rainbow?

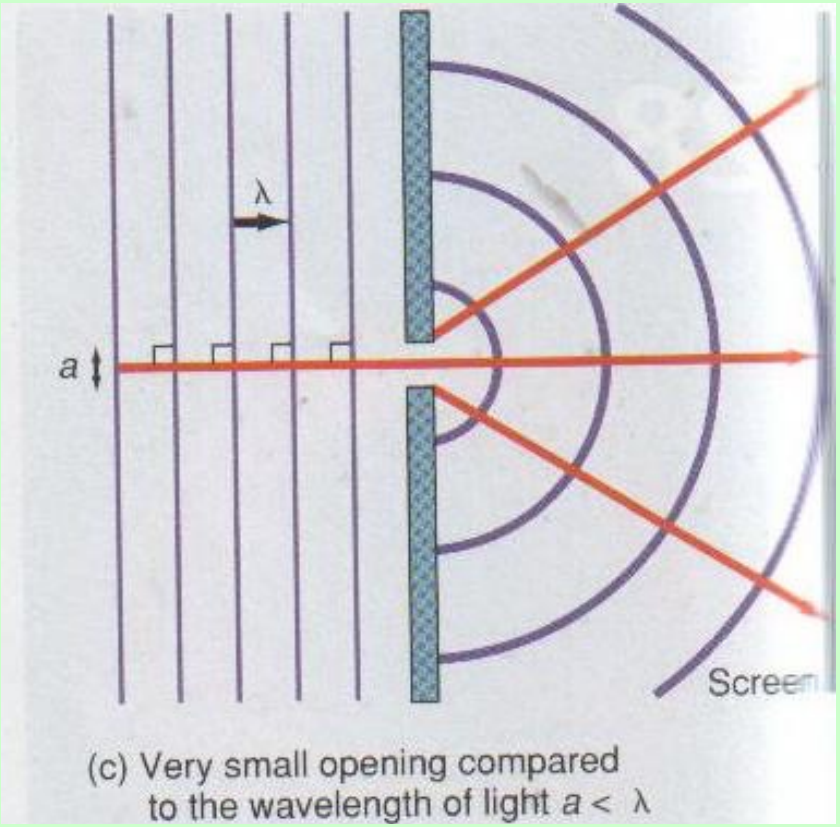
c. What is at m = 0?

Geometric Optics

Physical Optics



(a) Large opening compared to the wavelength of light $a >> \lambda$



(c) Very small opening compared to the wavelength of light $a < \lambda$

We can use rays

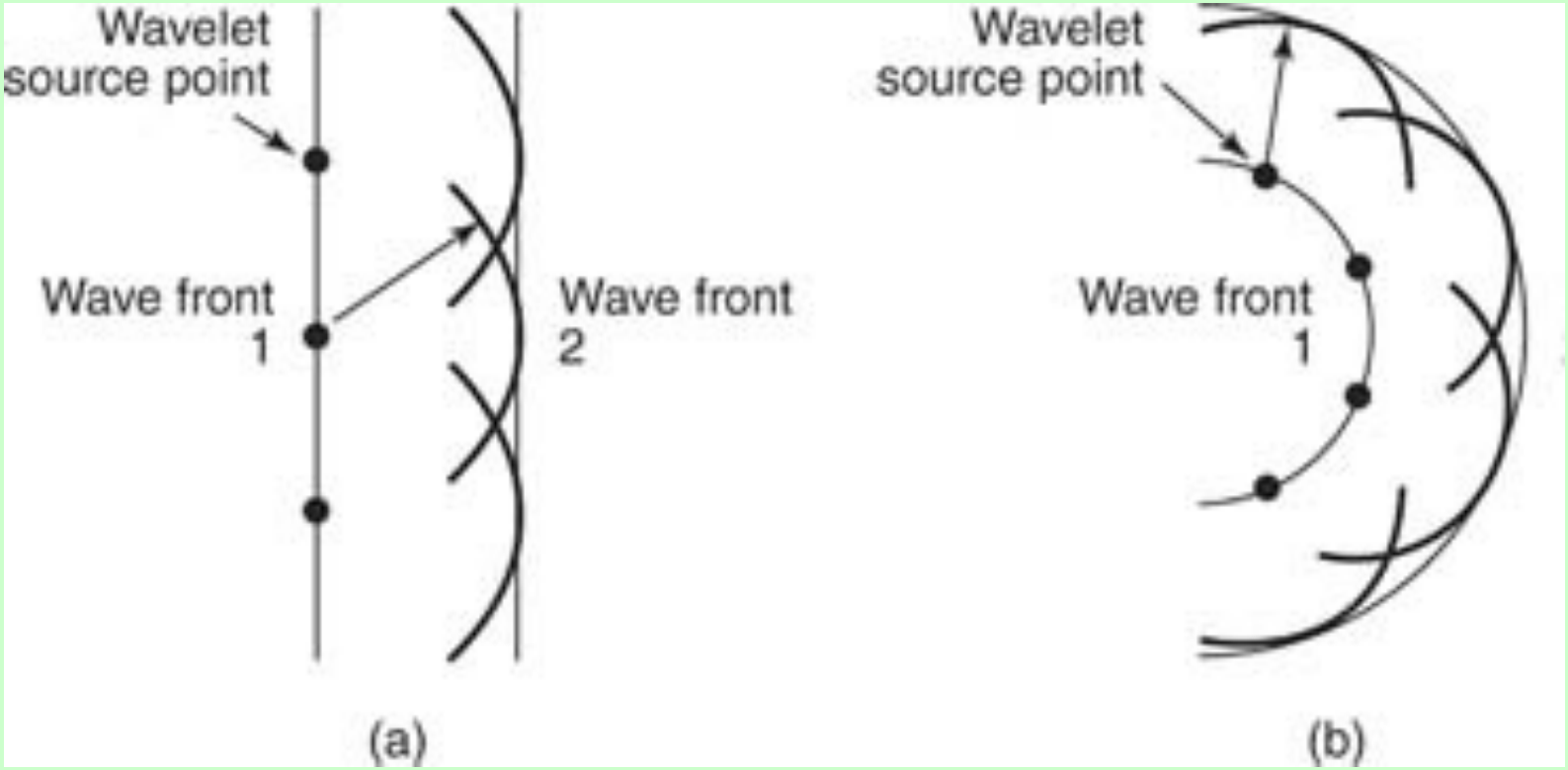
"particles"

We can't because **diffraction**

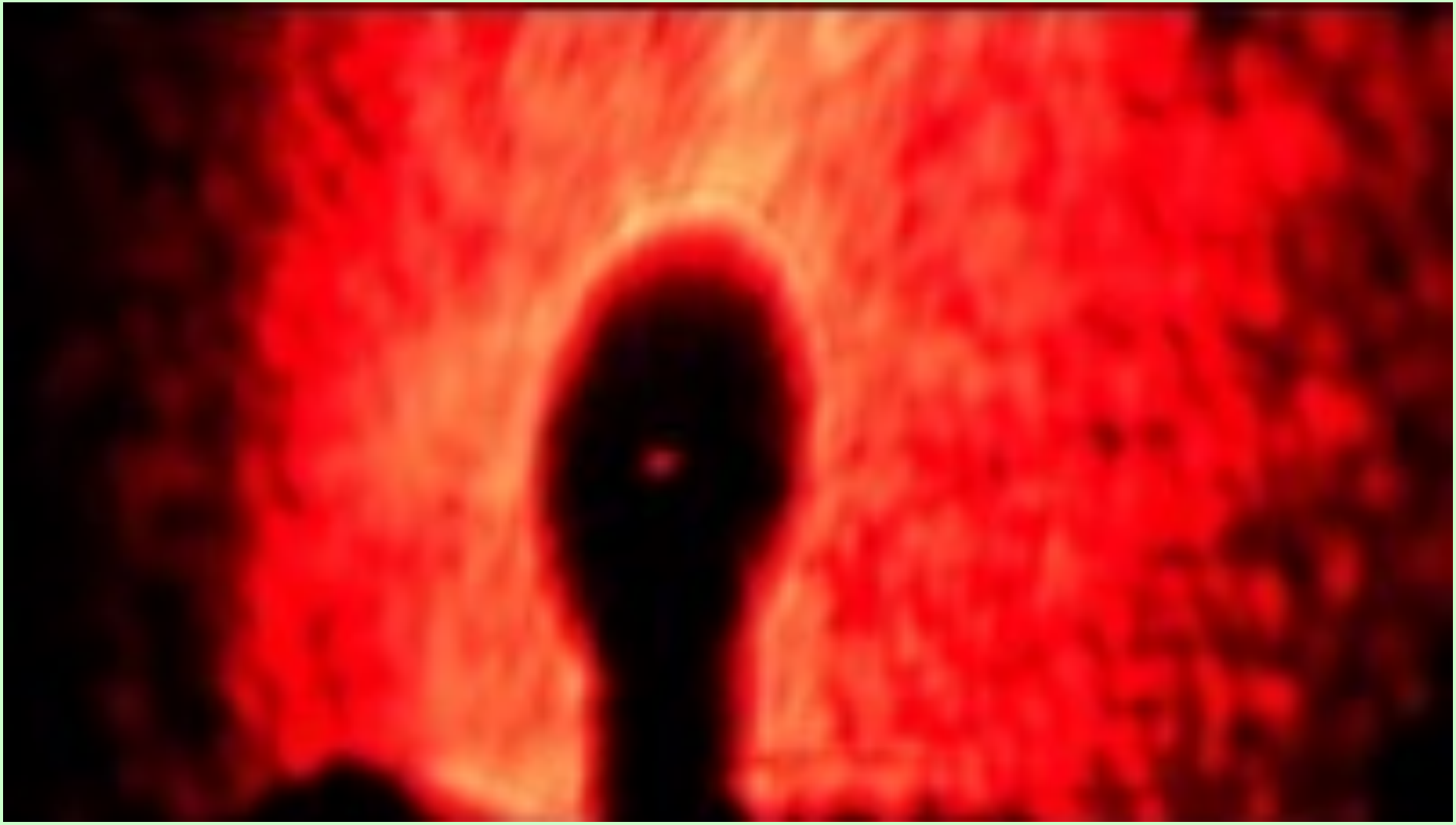
"waves"

Diffraction happens b/c each point can act as a Huygen's Source

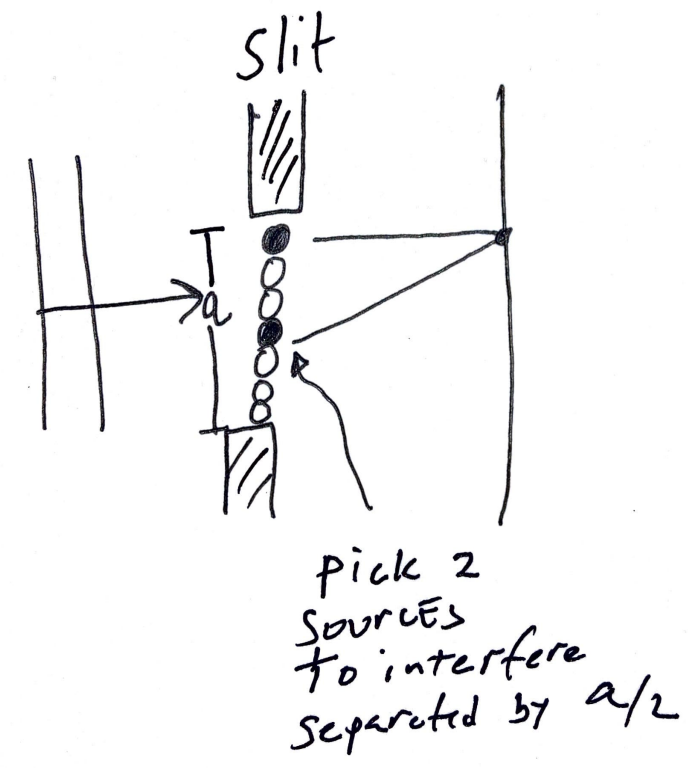
Huygen's Principle: each point on a wave front acts as a circular source of new wavefronts



Poisson Spot



Single Slit Diffraction:

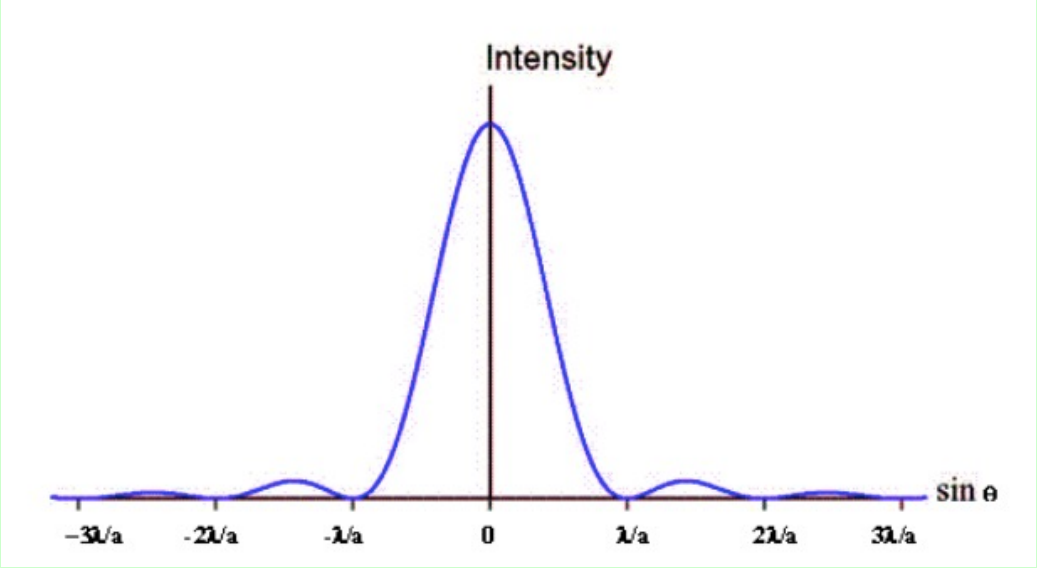


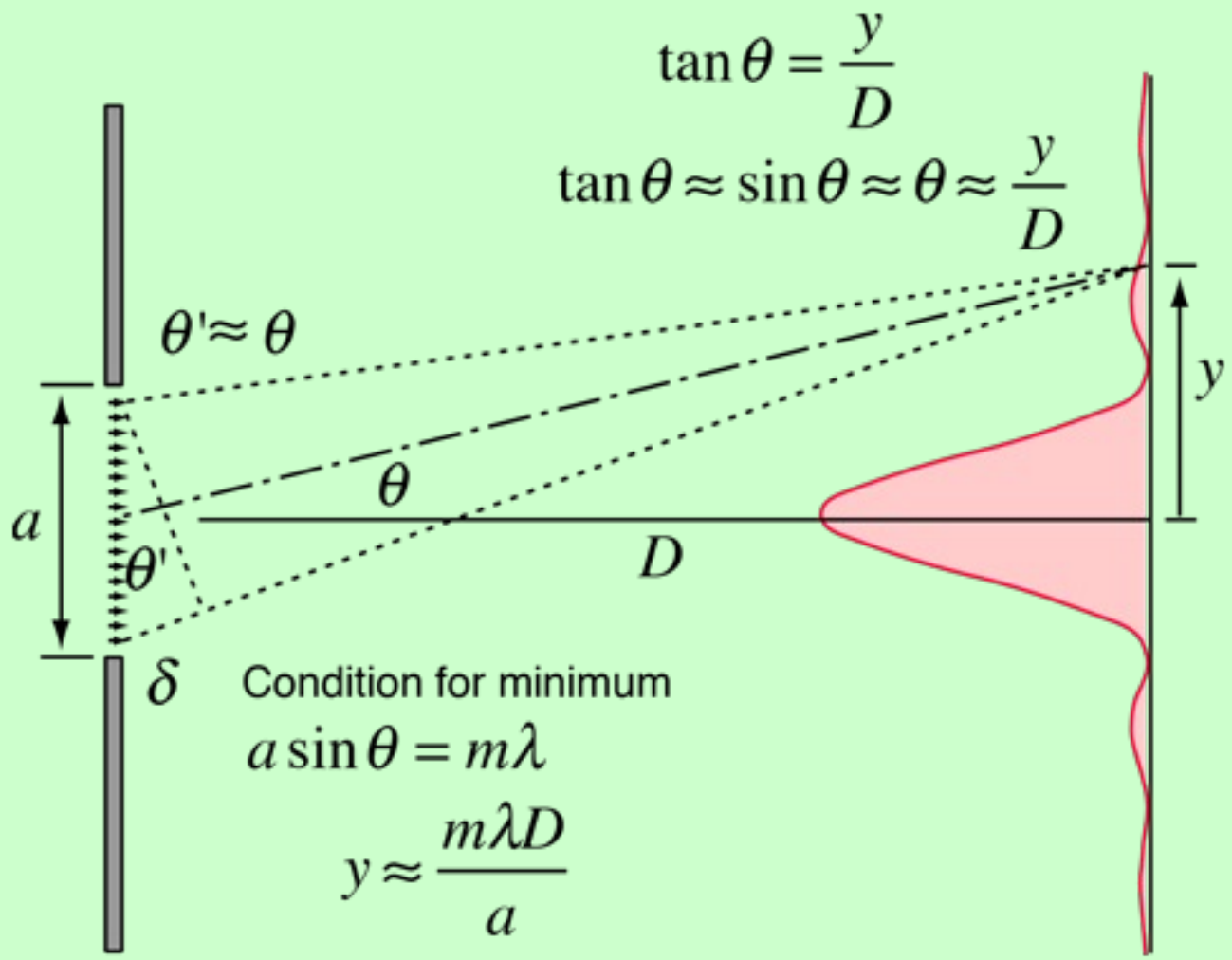
Where is the first minimum for two sources separated by a/2?

$$\frac{a}{2}\sin\theta = \frac{1}{2}\lambda$$

So the first **minimum** will happen at:

$$\sin\theta = \frac{\lambda}{a}$$




$\tan\theta = \frac{y}{D}$
$\tan\theta \approx \sin\theta \approx \theta \approx \frac{y}{D}$
$\theta' \approx \theta$
a
θ
θ'
D
y
δ
Condition for minimum
$a\sin\theta = m\lambda$
$y \approx \frac{m\lambda D}{a}$

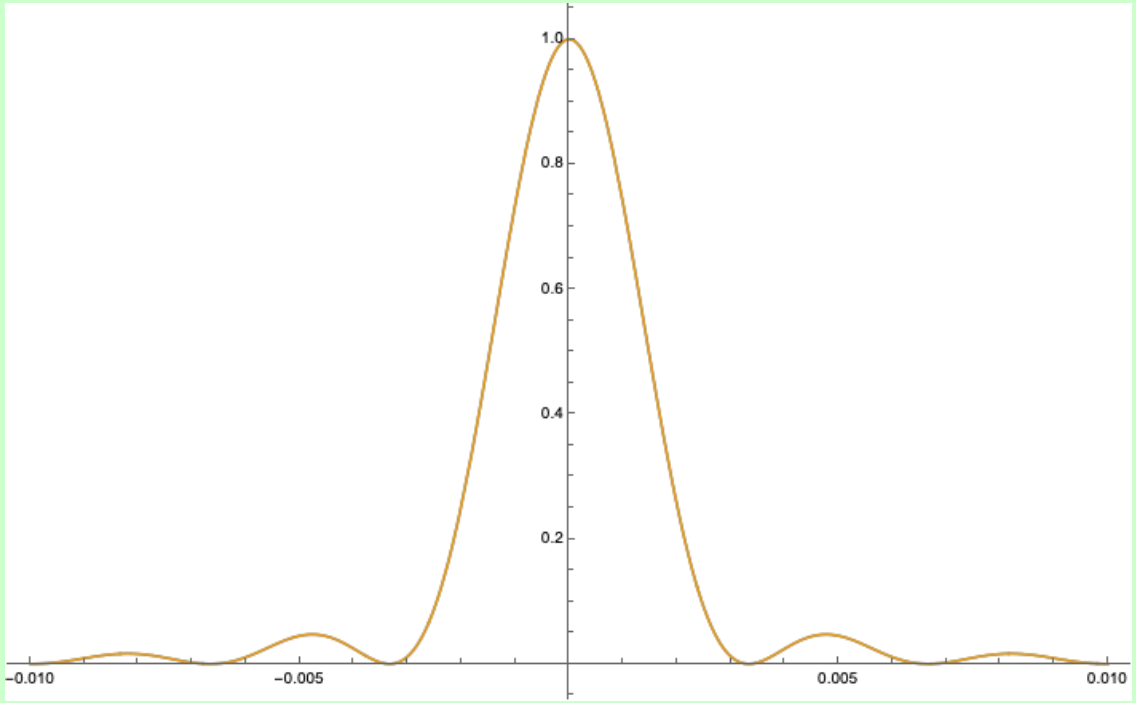
Ex. A single slit has a width of 15 microns. 800 nm light is incident upon it

A. If the screen is 20 cm away, what is the screen position of the second min?

The Single Slit Intensity Pattern

$$I = I_0 \frac{Sin^2(\beta/2)^2}{(\beta/2)^2} = \boxed{I_0 \; Sinc^2\,(\beta/2)}$$

Where $\beta = \frac{2\pi a \sin\theta}{\lambda}$



Ex. A single slit has a width of 150 microns. 500 nm light is incident upon it

A. Find the first position on the screen where $I = I_0/2$, assuming the screen is 2.5 m away. You will need to use Mathematica's Nsolve[]